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The comparison between single-point method and footprint-integrated validation method of the remote-sensing retrieval of evapotranspiration: a case study at Daman site

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ABSTRACT

In single-site validations, site-pixel, synthetic pixel and footprint-integrated methods are commonly used to validate evapotranspiration (ET) retrieval at the pixel scale. To reveal the reliability of these methods, this study analysed the performance of the three methods for ET retrieval validation based on eddy-covariance observations at the Daman site which is numerously used for validation in Heihe Watershed Allied Telemetry Experimental Research (HiWATER). The analysis at different spatial scales in different seasons showed that the validation results of the site-pixel method were more similar that of the footprint-integrated method than the synthetic pixel method. However, the validation results of the site-pixel method showed significant discrepancies from those of the footprint-integrated method, especially in summer at the 480 M scale which is near to the spatial resolution of MODIS with a large difference of more than 125.65 W M^{-2}. This error is due to aggregation and high spatial heterogeneity. The use of synthetic pixel can avoid this error. Normally, the error probably appears when the wind direction is 270°–360°, usually in summer, autumn and winter. In addition, the different aggregation methods had a slight effect on the ET estimation results. This study is helpful in determining the reliability of different single-point validation methods and provides a reliable way to select a validation method for single-site validation of remote sensing retrieval of ET.

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Evapotranspiration; input aggregation; nonparametric approach; upscaling; validation approach

SPATIAL HETEROGENEITY AGGREGATION

1. Introduction

AS an important component of the global energy and water cycle, ET, including evaporation from soil and transpiration from vegetation, is beneficial for researches on water resources and the thermal environment at the regional and global scale (Anderson et al. 2012; Z.-L. Li et al. 2009; Ma et al. 2018; Schotanus, Nieuwstadt

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and De Bruin 1983; Senay et al. 2011). Currently, many methods of remote sensing retrieval of ET have been proposed, such as the Surface Energy Balance System (SEBS), Surface Energy Balance Algorithms for Land (SEBAL) and triangle method (Bastiaanssen et al. 1998; Elhag et al. 2011; Gokmen et al. 2012; Ramoelo et al. 2014; Su 2002; Sun et al. 2011). Accordingly, various remote sensing ET products retrieved by these methods have been produced, such as MOD16, ET monitors, PML and PEW (Carlson 2007; Fu et al. 2022; Zhang et al. 2019; Zheng, Jia and Hu 2022). As one of the most reliable references for ET observations, single-site observations derived from EC have been used to validate the accuracy of these methods and products (Cheng et al. 2021; Kuppel et al. 2012; X. Li et al. 2021; Wilson et al. 2001; F. Xu et al. 2017). Usually, flux sites with EC are composed of a network to validate ET in different land covers and climates, such as FLUXNET, ChinaFlux and HiWATER (Baldocchi et al. 2001; X. Li et al. 2013; Velpuri et al. 2013; Yu et al. 2006).

Considering that the footprint of EC is believed to be the true coverage of EC observations, for the single-site validation of remote sensing retrieval of ET, the pixels located at the footprint of EC should be weight-averaged to validate the remote sensing ET (Aguirre-García et al. 2021; Ody et al. 2016; H. Wang et al. 2016). Limited by the complex process of footprint extraction, the pixel located at the EC site was always regarded as the 'true' pixel equivalent to the coverage of ground observations considering the homogeneity of land cover around the EC sites, especially for sites belonging to well-known networks (S. Liu et al. 2016). For example, the Daman site, one of the most famous sites in HiWATER, was widely chosen as the ground site for the reliable validation of the remote sensing ET owing to its relatively homogenous land cover (maize) and flat terrain (X. Li et al. 2018). At present, there are more than 500 studies about Daman site. Accordingly, single-site validations of the remote sensing retrieval of ET have been widely performed at the pixel located at the Daman site in numerous researches (Feng et al. 2023; X. Li et al. 2021; D. Wang et al. 2021). Although these validations are believed to be reliable, is the single-site validation of the remote sensing retrieval of evapotranspiration at the pixel scale reliable? Therefore, this study compared the validation results of site-pixel, synthetic pixel and footprint-integrated methods to investigate the reliability of the single-point validation method. Compared with the site-pixel and synthetic pixel methods (Frazier 2014), the footprint-integrated validation method can improve ET estimation and ground measurement at the spatial scale (Brunsell, Ham and Arnold 2011). In addition, the results of the single-point validation method may appear errors due to the inconsistent ET footprints under different atmospheric conditions.

To reveal the potential unreliability of the single-point validation method, the ET was retrieved by the RS-NP model, and the related pixels located at the Daman site of HiWATER were compared with EC observations at different spatial scales in different seasons. Then, the validation results of the site-pixel method, synthetic pixel method and footprint-integrated method were compared. The footprint-integrated method was used to validate the reliability of the single-site validation approach under various circumstances. The effect of atmospheric conditions on footprint-integrated validation was also investigated.

2. Study area and data

2.1. Study area and site

The study area (Figure 1) is located in Zhangye City, which is in the middle reaches of the Heihe River Basin (99°57′-100°59′E and 38°22′-39°25′N). This area includes oases and deserts. The Heihe River Basin is the second largest inland basin in Northwest China. The study area has a temperate continental climate with an average temperature of 6–8°C and an annual precipitation of 115.6 mm. Daman is a part of the Heihe Watershed Allied Telemetry Experimental Research (HiWATER) experiment. This experiment used EC systems and automated weather stations (AWSs) to measure the required data of the sites such as the LE. The measured data were intercompared and verified in May 2012, so the observed data were reliable. This study used Daman to assess the retrieved ET. The underlying surface of Daman is a cropland. The longitude of Daman is 100°22′19.91′, and the latitude is 38°51′18.36′.

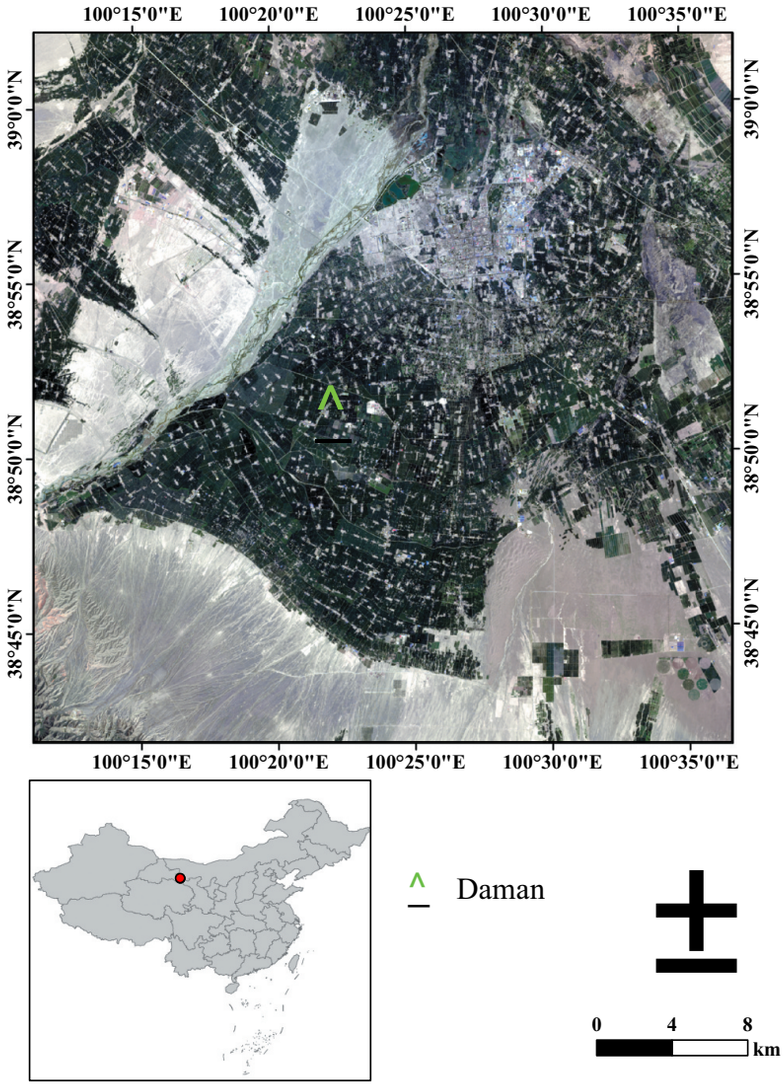


Figure 1. Distribution of study area and location of Daman.

2.2. Site observation data

The observation data of the site included the sensible heat flux (Hs), latent heat flux (LE), soil heat flux (Gs), soil moisture (Ms) and soil temperature (Ts). Hs and LE are measured by the EC, and GS, Ms and Ts are derived by AWSs. However, according to Burba G (Burba and Anderson 2010), the flux data measured by the EC have a problem. However, the energy closure rate of the flux data is relatively low. Therefore, the observation data of HiWATER must be revised using the residual energy term method. The LE calibrated using this method can be used to verify the ET retrieved from the RS-NP model.

2.3. Remote sensing and reanalysis data

In this study, Landsat 8 Level 2(L2), ERA5-Land and the China Meteorological Administration Land Data Assimilation System (CLDAS) products were used as the input data for the RS-NP model to retrieve ET. The time of input data is 4:00 (UTC). The data of Landsat 8 is remote sensing data, and it is available at the EarthExplorer website (https://earthexplorer.usgs.gov). Landsat 8 L2 has eight bands including bands 1 through 7 and band10. This study selected 13 January, 18 March, 9 August and 26 September to represent the winter, spring, summer and autumn of 2014, respectively. The spatial resolution of Landsat 8 L2 is 30 m. ERA5-Land and CLDAS are reanalysis data. The temporal resolution of ERA5-Land and CLDAS is hourly. The spatial resolution of ERA5-Land and CLDAS are $0.1^\circ \times 0.1^\circ$ and $0.0625^\circ \times 0.0625^\circ$, respectively. The RS-NP used the 2 m dewpoint temperature product of ERA5-Land. The CLDAS is produced by the China National Meteorological Center. CLDAS data include atmospheric temperature (TMP), surface pressure (PRS) and specific humidity (RHU).

3. Methods

3.1. ET estimation

This study used the RS-NP model (Pan et al. 2022) to estimate the ET (Figure 2). First, we aggregated the remote sensing data including ERA5-Land and CLDAS data to the fine spatial resolution of Landsat 8 data (30 M). These remote sensing data (30 M scale) were then used to calculate the ET in the RS-NP model. Thus, ET estimation at the 30 M scale was calculated. Second, the input data of the RS-NP model, including Landsat 8, ERA5-Land and CLDAS data were aggregated to subsequent spatial resolutions. The aggregation resolutions used in this study were 120 M, 240 M, 480 M and 960 M scale. The nearest neighbour method was used for aggregation. Finally, we calculated the ET estimation result after aggregating it to different scales on the RS-NP model.

3.2. Footprint model

The flux footprint can be used to confirm the flux source area of the ground measurement and the relative flux contribution to the area. This study used Kljun's Flux Footprint

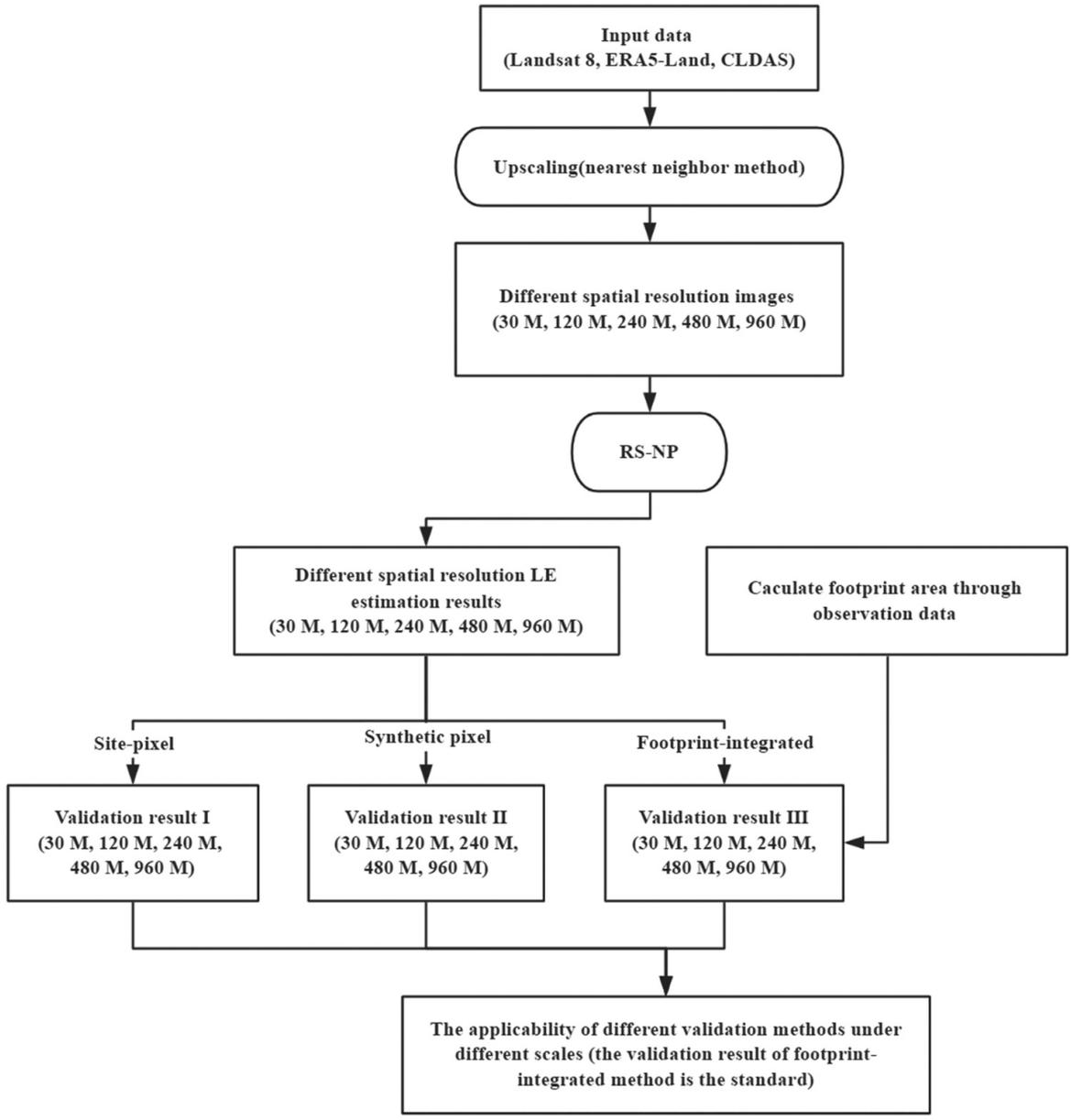


Figure 2. The flowchart of ET estimation and validation.

Prediction (FFP) (Kljun et al. 2015; Kljun, Rotach and Schmid 2002) to calculate the half-hourly flux footprint of EC measurement. This flux footprint model not only provides the extent of footprint estimates but also considers the effects of surface roughness length. The footprint function can be expressed by crosswind-integrated footprint $\bar{f}^{y}(x,y)$and a Gaussian crosswind concentration distribution function $D_y(x,y)$ (Horst and Weil 1992). The footprint function is expressed as follows:

$$
\begin{aligned}
f(x,y,z_m) &= D_y(x,y) \times \bar{f}^{y}(x,z_m) \\
D_y(x,y) &= \frac{1}{\sqrt{2\pi}\sigma_y} \times \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \\
\bar{f}^{y}(x,z_m) &= -\frac{K_c}{U} \times \frac{\partial D_z}{\partial z}
\end{aligned}
\tag{1}
$$

where x, y and z are streamwise direction, crosswind location and height, σ_y is commonly expressed as a function of atmospheric stability and distance downwind of the source, K_c is an eddy diffusivity, U is the effective speed of plume advection in x, D_z is the vertical concentration distribution function.

The input data of the FFP model can be obtained from EC measurements including the height above ground (zm), displacement height (d), Obukhov length (L) and boundary layer height (h). The boundary layer height data were obtained from the ERA5 product. The flux contribution to the source area was set to 90%. The flux source area was then obtained and used to validate the ET estimation. The specific method calculates the average value of the pixels contained in this area and uses it as validation value.

3.3. Site-pixel method and synthetic pixel method

In the single-site validation of the remote sensing retrieval of ET, the validation result of the site-pixel method is based on the value of the pixel where the site is located.

The synthetic pixel method is based on input aggregation which scales remote sensing data to different resolutions, including 120 M, 240 M, 480 M and 960 M scale. The specific procedure of the input aggregation is to select the corresponding number of 30 M resolution pixels around the site to be validated. The distribution of these pixels is located around the Daman site. Subsequently, a single corresponding resolution pixel can be obtained using the nearest neighbour method (NN). Finally, remote sensing data were used to retrieve the ET on the RS-NP model after input aggregation, and the ET estimation of the RS-NP model was the validation result of the synthetic pixel method. The area-weighting aggregation principle of the synthetic pixel method is defined as follows:

$$Y = \sum_{k=1}^{n} X_k \times \frac{S_k}{S} \tag{2}$$

where n is the number of selected pixels, X_k and S_k are the value and area of the k_{th} pixel, S is the area of the aggregated pixel. Figure 3 illustrates the site-pixel method and synthetic pixel method.

3.4. Evaluation method of spatial heterogeneity

The spatial heterogeneity of ET was evaluated using coefficient of variation (CV). The CV can be regarded as an assessment index for spatial heterogeneity based on the ET retrieved from the RS-NP model. CV is defined as the ratio of the standard deviation to the mean. The CV was calculated as follows:

$$CV = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2}}{\bar{x}} \tag{3}$$

where n is the number of the pixels, x_i are the i_{th} ET value, $\bar{x}$ is the average ET value. The spatial heterogeneity was greater when the CV was greater.

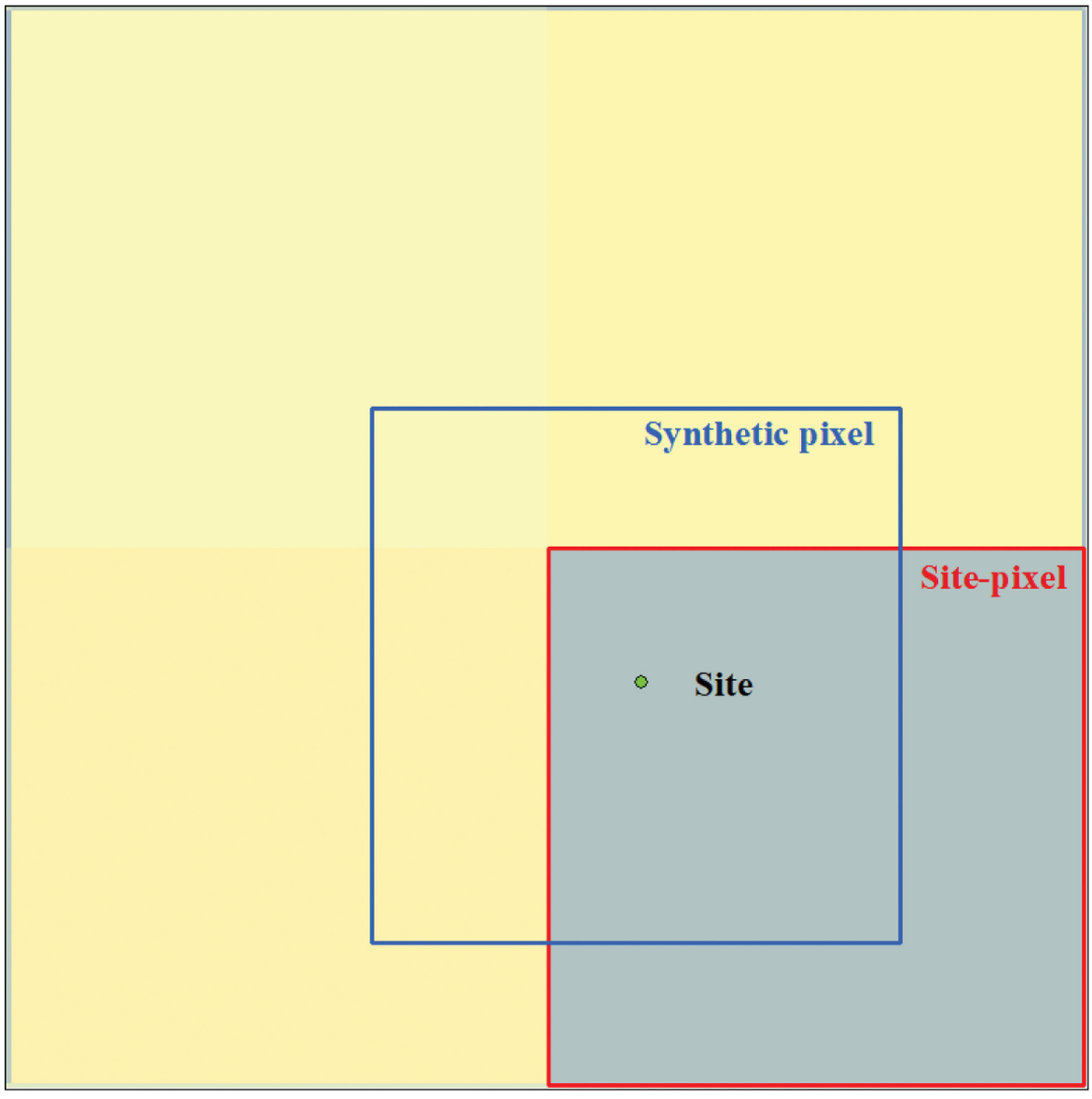


Figure 3. The illustration of (*a*) site-pixel method and (*b*) synthetic pixel method. The square boxes represent the validation area of different methods.

3.5. Metric for accuracy assessment

This study chose the RE to evaluate the accuracy of ET estimation, which can be described as follows.

$$Bias = \frac{\sum_{i=1}^{n} (e_i - o_i)}{n} \tag{4}$$

$$RE = \frac{|Bias|}{\frac{1}{n}\sum_{i=1}^{n} o_i} \times 100\% \tag{5}$$

where e_i is a retrieved value and o_i is a reference value.

Table 1. CV of the area around Daman and the entire study area in the four seasons.

	Area around Daman				Whole study area			
Scale	Winter	Spring	Summer	Autumn	Winter	Spring	Summer	Autumn
30M	0.11429	0.06535	0.10768	0.08183	0.66664	0.49759	0.69327	0.69526
120M	0.08889	0.03742	0.06542	0.04469	0.66647	0.49684	0.69317	0.69517
240M	0.10586	0.08089	0.06167	0.05751	0.66740	0.49554	0.69288	0.69496
480M	0.18135	0.02939	0.17896	0.10390	0.66569	0.49444	0.69137	0.69522
960M	0.05649	0.06268	0.00121	0.00282	0.66227	0.49249	0.69081	0.69458

4. Results and discussion

4.1. The spatial heterogeneity of sites and whole study area

To reveal the ET spatial heterogeneity of the area around Daman, the CV within was calculated at different spatial resolutions in the area around Daman and the entire study area in the four seasons (as shown in Table 1). We found that the CV at the 480 M scale was higher than that at other scales in all seasons except spring in the area around Daman. In addition, spatial heterogeneity within the entire study area was estimated. Generally, CV should be higher if the scale is higher except for 480 M scale. This exception was possibly related to the area with imperious surface in the southeast direction of the Daman site. The size of that area is about 450 M × 250 M. It was relatively large in the aggregate area of 480 M. However, the area was relatively small in the aggregate area of 960 M. Therefore, the CV at 480 M scale was significantly larger than that at 960 M scale. For the CV of the entire study area, it did not change significantly during the four seasons when aggregated to different resolutions. Even though the CV of Daman is not high in the entire study area, which means that the general heterogeneity is relatively satisfactory at all spatial scales according to, the CV increased at the 480 M scale around the Daman site in the summer. For example, the CV of Daman at the 480 M scale in summer was 0.17, and it was only 0.06, 0.06 and 0.001 at the 120 M, 240 M and 960 M scale, respectively. This indicated high spatial heterogeneity around the Daman site at the 480 M scale in winter, summer and autumn.

4.2. Validation of remote sensing ET

In order to investigate the reliability of the single-point validation method and the cases where errors will occur, we compared the validation results of different methods. Figure 4 shows observation and validation results of Daman when using different methods, including the site-pixel method, footprint-integrated method and synthetic pixel method. When using the site-pixel method, the validation result of ET changed significantly at the 480 M scale in all seasons except for spring, especially in summer. The variations in ET estimation can exceed 134.33 W M^{-2}. This indicated that this validation method might be incorrect at the 480 M scale, which is near the spatial resolution of MODIS, especially in the summer.

The validation result of the synthetic pixel method indicated that this method is almost unaffected by the different spatial resolutions of ET estimation in the four seasons. The discrepancies in ET varied within 4 W M^{-2}. The validation results of the footprint-integrated method changed insignificantly during the four seasons. The discrepancies

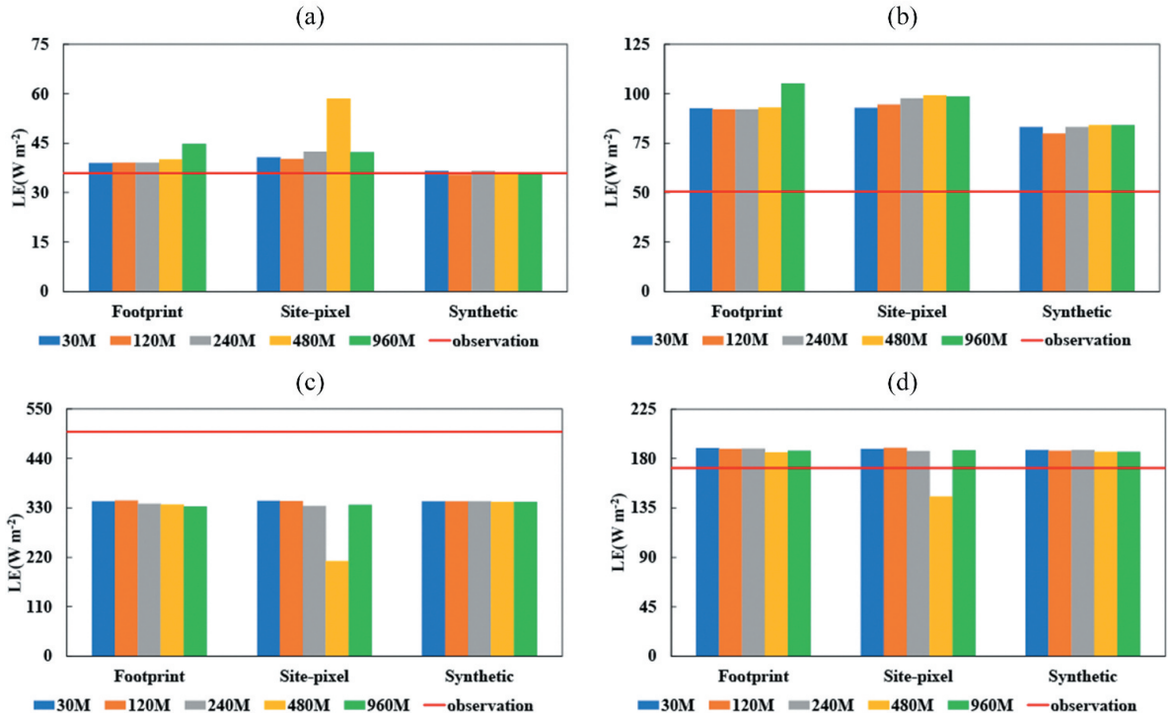


Figure 4. Comparison of observations and validation results at Daman using site-pixel, footprint-integrated and synthetic pixel methods at different resolutions in four seasons (a. Winter, b. Spring, c. Summer, d. Autumn).

in ET varied within 12 W M^{-2} when the footprint-integrated method was used. The difference in validation results between the site-pixel method and the footprint-integrated method can be up to 125.65 W M^{-2}.

The RE of the three methods were calculated based on the validation result and the LE observation data. The RE of ET estimation at Daman using different validation methods in four seasons at different scales is illustrated in Figure 5(a-c). We found that the variations

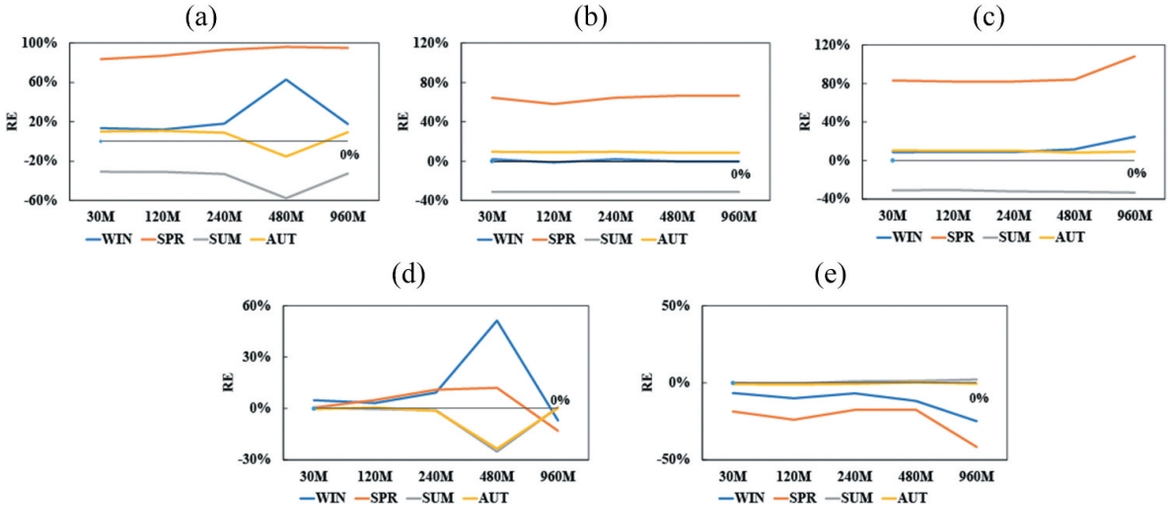


Figure 5. Variation in RE at Daman under different validation methods (a. site-pixel method, b. synthetic pixel method, c. footprint-integrated method) and the RE difference between single-point method and footprint-integrated method (d. site-pixel method and footprint-integrated method, e. synthetic pixel method and footprint-integrated method).

in the RE of ET estimation when using the site-pixel method were less than 13% in spring. However, in the other three seasons, the variation in RE could be up to 49.4% when aggregated to different scales. The RE of the synthetic pixel method remained stable, with a variation of less than 6%. The variations in the RE of the footprint-integrated system were in four seasons. The largest change in variation was 25% when the scale was 960 M. Figure 5(d,e) show the RE difference between the single-point method and the footprint-integrated method. The RE difference between the site-pixel method and the footprint-integrated method was approximately 0%, except at the 480 M scale. The difference between the synthetic pixel method and footprint-integrated method is approximately 0% in summer and autumn, while the difference can be up to 23.97% in winter and spring at different scales except 960 M. The trend of the difference was the same at different scales except at 960 M. The special circumstance is because of the small numerical changes in the footprint-integrated method at the 960 M scale. In short, the validation result of the site-pixel method is closer to the footprint-integrated method than the synthetic pixel method except at the 480 M scale.

Figure 6(a) shows the variation in the pixels around the Daman site at different spatial resolutions. Even though the homogeneity and terrain around the Daman site were believed to be reliable, the value of the selected pixel changed dramatically from 345.62 W M^{-2} to 211.29 W M^{-2} at the 480 M scale obviously. It seems that there are different land covers (differing from maize) around the site, and the distribution of the land cover affected the reliability of the validation at the 480 m-scale, especially in the summer. In this case, the values of pixels located southeast of Daman are lower (as shown in the first image of Figure 5), so the value of the aggregated pixels is also lower (the 480 M scale pixel in the southeast direction). When using the site-pixel method for validation, the site is located in this pixel when aggregated to 480 M scale. Therefore, the site-pixel method may result in errors under certain conditions.

Figure 6(b) shows the discrepancies between the selected pixels of the site-pixel method and the synthetic pixel method at Daman in the summer. The variations were less than 16 W M^{-2} at 120 M, 240 M and 960 M resolution. The discrepancy of the pixel in the southeast direction was approximately 131.86 W M^{-2} at 480 M resolution. This explains the error in the validation results of the site-pixel method at 480 M. The synthetic pixel method is more reliable at the 480 M scale because the pixels that cause the wrong 480 M pixel are relatively far from the location of Daman and the synthetic pixel method does not include these pixels in the calculation.

In summary, the validation result of the site-pixel method is closer to the footprint-integrated method than the synthetic pixel method except at the 480 M scale. However, the site-pixel method may result in errors due to aggregation and high spatial heterogeneity. The difference in the validation results between the synthetic pixel method and footprint-integrated method was slightly higher. However, the synthetic pixel method can better avoid this error because of the aggregation and high spatial heterogeneity at a certain scale.

4.3. *Influence of atmospheric conditions on footprint-integrated method*

The validation result of the footprint-integrated method was influenced by its flux source (Bai et al. 2015), which was determined by the atmospheric conditions. Therefore, it is

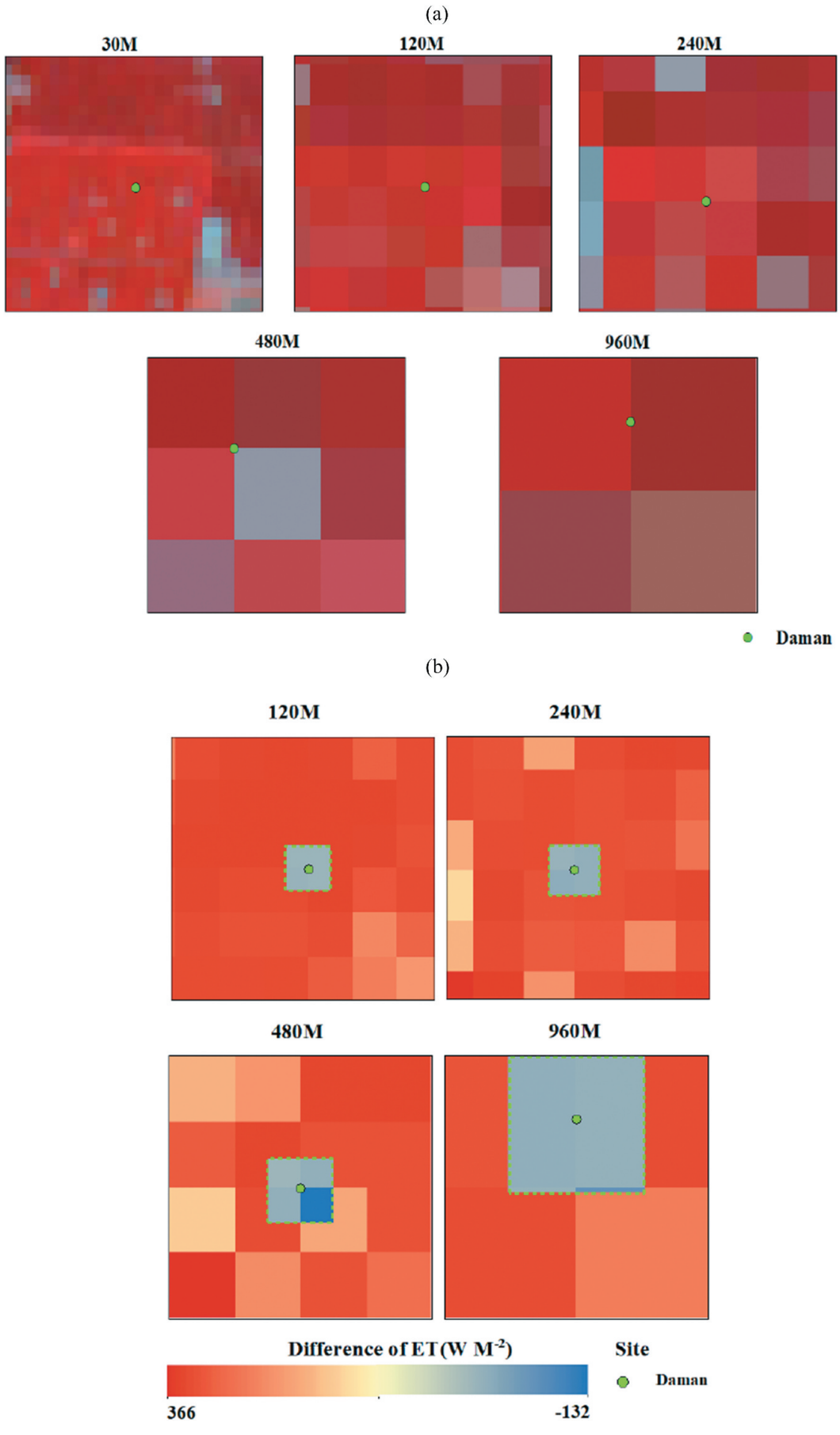


Figure 6. (a) The false color remote sensing image (composited by band 5, band 4 and band 3 of Landsat 8) around Daman site when aggregated to different spatial resolution in summer and (b) the difference of ET between the pixels of site-pixel method and synthetic pixel method at Daman in summer which are associated with the validation (the area of synthesized pixel is in the green box).

Table 2. Details of different atmospheric conditions at the Daman site.

EC	Wind direction (°)	Wind Speed (M s^{-1})	Sigma_v (M s^{-1})	Friction velocity (M s^{-1})
1	67.847	2.509	0.657	0.221
2	154.600	1.316	0.417	0.117
3	212.974	3.513	0.810	0.236
4	234.379	2.907	0.522	0.422
5	278.144	4.760	1.762	0.553
6	345.041	3.478	1.016	0.331

necessary to investigate the relationship between variable atmospheric conditions and the direction, size and shape of the flux source area. We can then determine the circumstances under which the validation results of the site-pixel method may show significant discrepancies compared with those of the footprint-integrated method. To better understand the effects of atmospheric conditions on the results of the footprint-integrated method, six atmospheric conditions were selected for the summer of 2014 at the Daman site (Table 2). These conditions were classified into six EC scenarios. Finally, we obtained validation results for ET at different scales in summer.

The flux source area varied because of different atmospheric conditions (Figure 7). We found that the flux source area was mainly influenced by the wind direction, wind speed, and friction velocity. Wind direction determines the orientation of the source area. The flux area of the footprint is directly affected by wind speed and friction velocity (Kljun et al. 2015). When other atmospheric conditions were the same, the wind speed was higher, and the area was longer. In addition, the area became larger and wider as the friction velocity decreased.

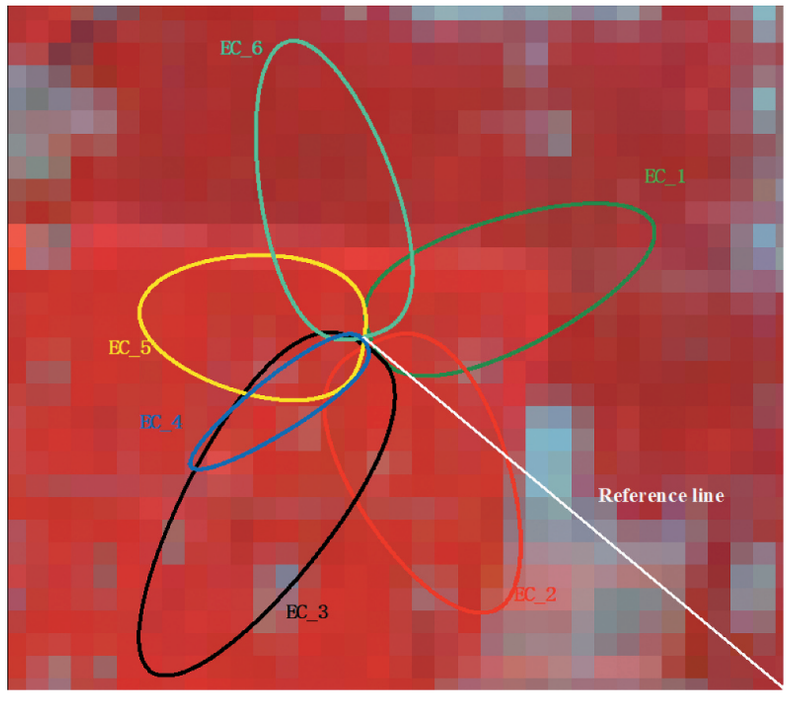


Figure 7. Distribution of footprints obtained by EC_1 through EC_6 at the Daman site and the resolution of the background image is the native resolution of Landsat 8. The direction of the special pixels located near the reference line was approximately 135°.

Table 3. ET estimation based on EC_1 through EC_6 and the validation results of site-pixel and synthetic pixel methods at different scales in summer.

	Different validation results of ET (W M^{-2})							
Scale	1	2	3	4	5	6	Site-pixel	Synthetic
30M	339.57	329.06	336.59	336.56	344.64	345.68	345.62	344.10
120M	339.74	305.69	336.69	337.09	344.56	346.76	344.96	344.18
240M	334.49	333.88	327.74	333.92	338.26	342.14	333.88	344.10
480M	296.67	292.10	296.67	296.67	339.35	336.95	211.30	343.16
960M	332.71	332.71	332.71	332.71	328.70	332.71	336.72	343.16

Table 3 shows the results of different situations according to the corresponding flux source area and the results of the site-pixel and synthetic pixel methods. There are some pixels whose flux values are different from those of the surrounding pixels (located in the southeast direction of Figure 7) at Daman. The direction of these pixels is marked as the reference line (135°). The results for EC_1, EC_2, EC_3 and EC_4 showed errors at the 480 M scale and the difference could be up to 42.9 W M^{-2}. The difference between the validation results of the site-pixel method and the results of EC_5 and EC_6 was at least 125.65 W M^{-2}. This was mainly influenced by the wind direction. The reason for this error is that the flux source areas of EC_1 to EC_4 include the 480 M scale pixel which is aggregated by the 30 M scale pixels in the southeast direction (defined as 135°). The difference between the validation results of the synthetic pixel method and footprint-integrated method was less than 14.46 W M^{-2} when the wind direction was 270° to 360° (EC_5 and EC_6). When the difference in direction between EC_4 and the pixels whose flux values are different from the surrounding pixels is approximately 90°, the validation result of EC_4 also shows an error even though its area is narrow and small.

The reason for the error at the 480 M scale is the aggregated 480 M scale pixel located in the southeast direction. The validation results appear to have errors, except when the wind direction is 270°–360° (EC_5 and EC_6). Figure 8 shows the distribution of the UTC-4:00 wind direction at Daman during the four seasons. This indicates that 270°–360° is one of the prevailing wind directions at Daman during the four seasons. Therefore, EC_5 and EC_6 often occurred. The result of the footprint-integrated method was true, and the validation results of the site-pixel method showed significant discrepancies compared with the footprint-integrated method when the wind direction was from 270° to 360°. Meanwhile, the validation results of the synthetic pixel method and footprint-integrated method were similar. When the wind direction was not from 270° to 360°, the validation result of the footprint-integrated method showed an error at the 480 M scale compared to other scales.

4.4. Effects of different aggregation methods

In remote sensing ET calculations, the aggregation of fluxes from high to low resolution is a common practice. Therefore, it is necessary to understand how different aggregation methods affect the ET retrieval results. To reveal the influence of pixel aggregation approaches on validation, this study used the input data of the synthetic pixel method and aggregated it to different scales in the nearest neighbour (NN), simple average (SA), bilinear (BL) and bicubic (BC) methods. The discrepancies in ET estimation among the

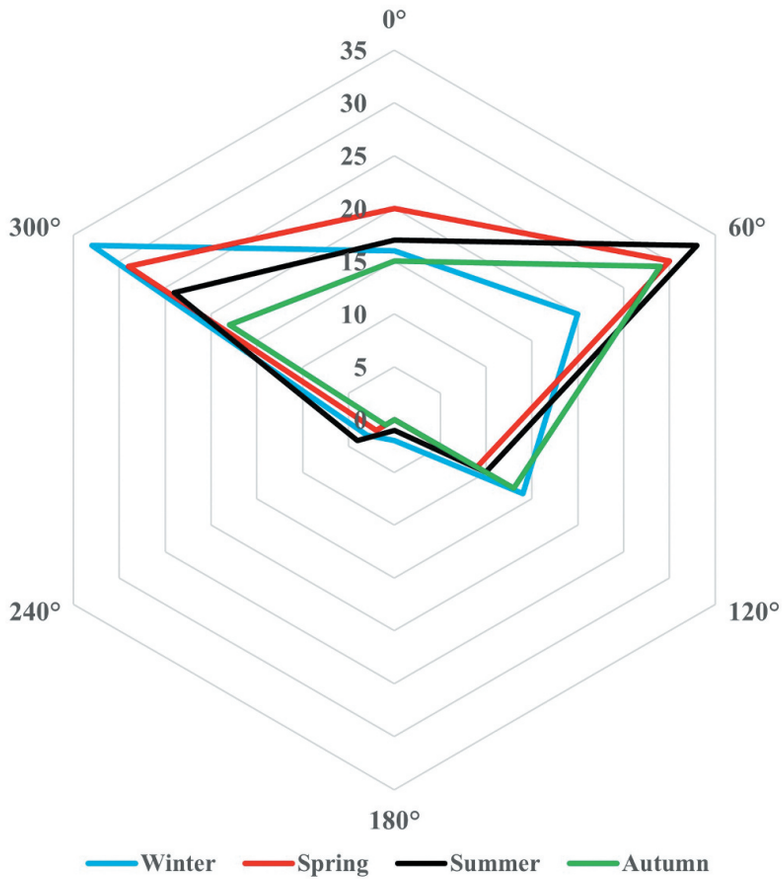


Figure 8. Distribution map of the UTC-4:00 wind direction at Daman during the four seasons (the unit of the vertical axis is the number of days).

pixels composed using different approaches are revealed at the different spatial resolutions in Table 4. Generally, at the same spatial scale, the discrepancies for different aggregation approaches varied within 8 W M^{-2} in all seasons at Daman. However, it seems that the simple average approach obviously affected the values of the synthesized pixels at the 960 M scale which is similar with the spatial resolution of MODIS, especially at the Daman site in the summer and autumn, with discrepancies of approximately 23 W M^{-2}.

To investigate the influence of different aggregation methods on the site-pixel validation method, we aggregated the input remote sensing data of Daman in summer to different scales using different methods including NN, SA, BL and BC methods (see Figure 9). The main difference in the results of the four methods is the pixels in the southeast at the 480 M scale, and the values of these pixels were 238.03 W M^{-2}, 298.87 W M^{-2}, 224.93 W M^{-2} and 223.69 W M^{-2}, respectively. Figure 10 shows the difference between SA and the other three aggregation methods. This indicates that the difference between the four pixels around the site is huge at 480 M.

In summary, the different aggregation methods had little influence on the validation results. The accuracy of the SA method was less than that of the other methods when the validation results showed errors due to spatial heterogeneity. This conclusion is similar with those of previous studies (Ershadi et al. 2013; Hong, Hendrickx and Borchers 2009). Ershadi et al. (2013) found that all aggregation methods yield a similar image-scale mean

Table 4. The ET estimation (W M^{-2}) of the Daman site after using different methods for input aggregation.

	Scale	NN	BL	BC	SA
Winter	30M	36.6188			
	120M	35.4452	36.1572	36.0017	36.8136
	240M	36.6188	36.1503	36.0305	35.6157
	480M	35.8310	36.1492	36.0865	36.9155
	960M	35.8310	36.1585	36.0865	38.7069
Spring	30M	83.1752			
	120M	79.9333	81.9055	81.4391	84.1506
	240M	83.1752	81.9063	81.5319	82.8822
	480M	84.1670	83.3766	83.3422	85.4854
	960M	84.1670	83.3863	83.3422	88.9847
Summer	30M	344.0965			
	120M	344.1778	344.4478	344.5084	343.3899
	240M	344.0965	344.4405	344.5114	342.2511
	480M	344.1576	343.8409	343.8319	336.1482
	960M	344.1576	343.8363	343.8319	321.6744
Autumn	30M	187.7727			
	120M	187.8715	186.3638	186.1118	187.3555
	240M	187.7727	186.3620	186.1772	188.1170
	480M	185.9820	185.9279	185.8935	187.1105
	960M	185.9820	185.9272	185.8935	180.5875

(mean of all pixels of the image) compared to the fine resolution ET images in terms of statistical metrics. On the other hand, it was found that the NN aggregated image images showed an improved match with their corresponding fine resolution images. In addition, the volumetric ET errors due to aggregation were calculated. The SA method did not produce errors, and errors were evident in the NN method maps for all the days. Compared to NN, the volumetric ET errors for BL and BC were lower. The influence of different aggregation methods may differ for various ET calculation models.

4.5. Discussion

In the researches of remote sensing image, it is generally considered that the information content of a pixel originates solely from within its area. However, a substantial portion of the spectral signal of each pixel comes from surrounding areas exist in an actual remotely sensed image. This is a result of many factors including atmospheric effects and image resampling (Forster and Best 1994; Markham 1985). Point spread function (PSF) is used to describe and characterize the above effects. PSF influences the accuracy of deriving surface information from satellite images on a per-pixel basis (Huang et al. 2002). The effect of PSF impacts the accuracy and reliability of downscaling (Q. Wang, Tang and Atkinson 2020). It is obvious that this impact also exists during the upscaling of remote sensing images. Therefore, PSF can influence the accuracy of remote sensing and field-measured ET under different scales. The error of validation result of site-pixel method at the 480 M scale in this study may be due to PSF. That error may be avoided by the spatial subpixel analysis method to some degree (Kaiser and Schneider 2008). Therefore, it is necessary to study this impact of PSF on validation methods in the future.

Daman site is widely used to verify the validation of ET remote-sensing retrievals. Many of these studies use the site-pixel method, and these studies use the MODIS scale of 500 M input data (H. Liu, Yongmin and Aihua 2023; J. Xu et al. 2019; Yang et al. 2024; Yuan et al.

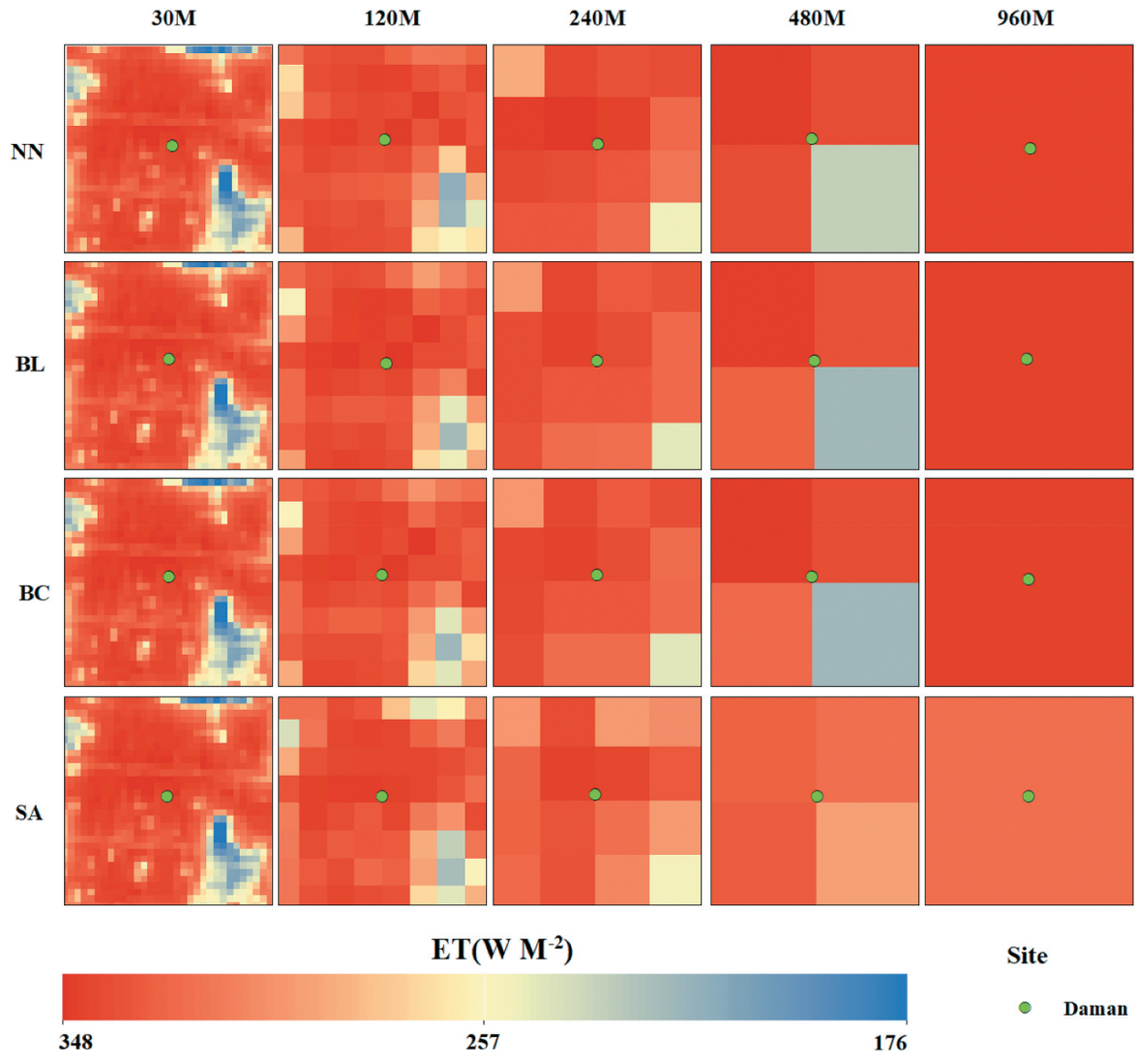


Figure 9. Spatial distribution of pixels using different interpolation aggregation methods.

2024; Zhuang et al. 2021). In addition, there are studies which have shown that the land surface heterogeneity could cause significant errors in ET estimation at 500 M scale (Tian et al. 2012). Therefore, according to our comparison results, the validation results of ET remote sensing retrieval may have errors in previous studies, and footprint-integrated method and synthetic pixel method are needed to validate those ET remote sensing retrieval results. In the future, we suggest that other researchers need to be careful when using Daman site for ET remote sensing retrieval of MODIS data at 500 M scale. We recommend to use footprint-integrated method and synthetic pixel method in the validation of ET retrieval.

5. Summary and conclusions

The aim of this study was to investigate the reliability of the single-point method under different conditions. This study compares the validation results of the site-pixel method, synthetic pixel method and footprint-integrated method based on the RS-NP model. We also examined the influence of different aggregation methods and atmospheric conditions on validation results. The conclusions are as follows:

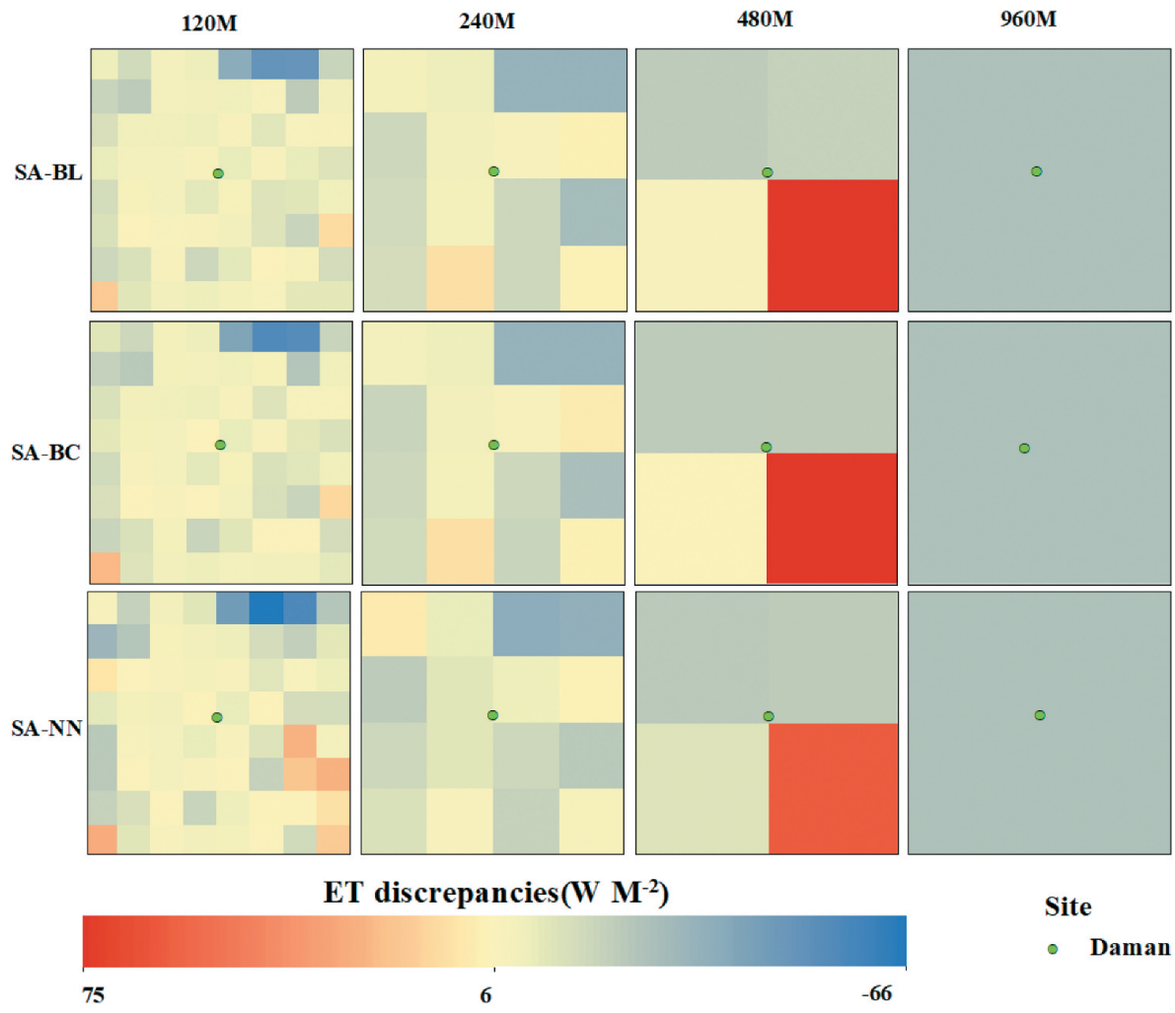


Figure 10. Difference in values between SA and the other three aggregation methods.

(1) When aggregating at different spatial resolutions, the ET spatial heterogeneity around Daman was higher at the 480 M scale. Meanwhile, the values of pixels located southeast of Daman site are lower compared with the other 480 M pixels around Daman site. Meanwhile, the Daman site is located on pixels with lower values of these. Therefore, we can confirm that the site-pixel validation is not always reliable.

(2) By comparing the validation results of the site-pixel, synthetic pixel and footprint-integrated methods, we find that the site-pixel method is generally closer to the footprint-integrated method. However, the validation result of site-pixel method may appear error compared to footprint-integrated method and the difference of ET value and RE can be up to 125.65 W M^{-2} and −25.17% in summer at 480 M scale. The synthetic pixel method can avoid this error although it is not closer to the footprint-integrated method than to the site-pixel method.

(3) The validation results of the site-pixel method showed significant discrepancies compared with the footprint-integrated method when the wind direction was 270°–360°. Under these circumstances, the difference between the synthetic pixel method and the footprint-integrated method is less than 14.46 W M^{-2}.

(4) The different aggregation methods including NN, SA, BL and BC, have only a slight effect on the ET estimation. The simple average (SA) method has

a relatively larger effect on the values of the synthesized pixels at the 960 M scale, especially at the Daman site in the summer and autumn, with discrepancies of approximately 30 W M^{-2}.

This study investigated the reliability of the single-point validation method in single-site validations of remote sensing retrieval of ET based on the footprint-integrated method. These results provide technical support for future studies on single-site validation. However, although the atmospheric conditions that make the validation results of the synthetic pixel method and footprint-integrated method are similar, it is still necessary to further explore the influence of atmospheric conditions on the reliability of the single-point method. In addition, it is also necessary to explore the impact of PSF on different validation methods.

Disclosure statement

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